

Option D – Energy and the Environment

10. (a) (i) The mean temperature of the Earth is 288 K. Assuming it behaves as a black body, show that the wavelength of the peak emission of radiation is in the infra-red region of the electromagnetic spectrum. [2]

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- (ii) Carbon dioxide, methane and water vapour are greenhouse gases. Describe the role these gases play in the greenhouse effect. [3]

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- (b) (i) The Greenland ice sheet is the second largest ice sheet in the world. Over the last two decades the mean decrease in volume of Greenland's ice is $3.0 \times 10^{11} \text{ m}^3 \text{ year}^{-1}$. Estimate the volume of water released into the surrounding ocean in this time period. [$\rho_{\text{ice}} = 920 \text{ kg m}^{-3}$, $\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$] [2]

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- (ii) Satellite images show that the edge of the Greenland ice sheet appears brown in places. Scientists report that this discolouration is due to microscopic plants flourishing in the melting ice. Explain the effect this could have on the rate at which the ice melts. [2]

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